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Willingness to Pay for "0 TFA" and the Induced Fat Health Halo: Evidence from China's Live-Streaming Food Market

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Abstract

Industrial trans-fatty acids (TFA) are a major diet-related risk factor. However, in China, disclosure on pre-packaged foods remains partial and is often communicated through voluntary "0 TFA" claims in e-commerce hero images on digital shopping platforms. These cues may generate a "health halo," distorting nutrient assessment. This study quantifies the willingness to pay (WTP) for bread and the induced fat-content "halo" using a daily panel of 59 top-selling bread SKUs from a major Chinese live-streaming platform. Estimation relies on a discrete choice framework applied to aggregate market data, testing robustness across specifications varying in coefficient heterogeneity and price endogeneity. The estimation results show no essential difference across specifications; all indicate that the "0 TFA" claim commands a positive WTP and induces a "health halo" for total fat. Taking the random coefficients model that addresses price endogeneity as an example, the claim commands a WTP of approximately 0.907 CNY per 100 g. Furthermore, it raises the marginal WTP for total fat from roughly 0.105 CNY to 0.232 CNY per 100 g, implying a halo valuation of about 0.127 CNY (≈121% increase). This occurs despite similar nutrient profiles, suggesting voluntary digital claims improve perceived safety while increasing tolerance for high-fat products.

Keywords: Trans-fatty acids; Health halo; Willingness to pay; E-commerce

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1. Introduction

Industrial trans-fatty acids (TFA) are unequivocally identified as a major diet-related risk factor for coronary heart disease. Meta-analyses and policy evaluations demonstrate that even low intakes of industrial TFA increase LDL cholesterol and cardiovascular risk, contributing to hundreds of thousands of deaths annually [1–3]. Consequently, numerous countries have limited TFA in foods or banned partially hydrogenated oils, while international organizations frame TFA elimination as a priority public health goal [4,5]. In 2018, the World Health Organization launched the REPLACE action package to eliminate industrially produced TFA globally [3]. These initiatives have established a consensus on the severe dangers of TFA, portraying it as a harmful lipid that should be strictly avoided [2].

While China has witnessed major progress in nutrition information disclosure through the National Food Safety Standard (GB 28050-2011), the regulatory architecture regarding TFA creates a specific information gap. The standard does not mandate universal TFA disclosure; manufacturers must list TFA content only if hydrogenated oils are

present or specific claims are made [6,7]. This insufficiency forces risk-averse consumers to rely on voluntary disclosure, turning “0 TFA” claims into strategic marketing tools [8–11]. This mechanism alters consumer willingness to pay (WTP) through two distinct channels. First, the claim itself acts as a direct value signal, increasing WTP by differentiating the product as "safe". Second, the claim might generate a “health halo”, which occurs when a single favourable nutrition or health cue leads consumers to exhibit a higher preference for other nutrients [12–15]. In our context, the prominently displayed “0 TFA” attribute may therefore act as a “license to indulge”, inducing consumers to increase the marginal utility of total fat by prioritizing taste over potential health concerns.

These effects are amplified in the online retail environment, where the nature of voluntary disclosure differs significantly from traditional retail [16–18]. In digital markets, voluntary disclosure is not merely a physical front-of-package (FOP) label but is primarily embedded in the hero image—the main product thumbnail displayed on search results and product pages [19]. In online shopping, the hero image serves as the primary and most salient visual touchpoint within a visually intensive environment where accessing back-of-pack NIP data entails high search effort [20]. This digital prominence amplifies the strategic impact of the disclosure: it not only commands a significant premium for the claim itself but also intensifies the health halo effect, potentially distorting the evaluation of high total fat more effectively [21].

Therefore, our research focuses on two central questions: (i) What is consumers’ WTP for a “0 TFA” claim itself in an online market? and (ii) Does the presence of this claim in the hero image create a “fat health halo” that distorts the marginal utility of total fat?

To address these inquiries, this study focuses on the sliced bread category within a major Chinese live-streaming e-commerce platform. The analysis utilizes a comprehensive daily product-level panel dataset that links hero-image “0 TFA” claims with back-of-pack nutrient contents, prices, and sales. Estimation relies on a discrete choice model using data aggregated at the market level [22–24]. This instrumental variable logit (IV-Logit) framework addresses price endogeneity using aggregated market-level data, thereby enabling the structural decomposition of consumer utility to isolate the specific WTP for the “0 TFA” claim and quantify how its presence shifts the marginal utility of total fat.

Our study yields three main contributions. First, we provide the first revealed-preference estimates of WTP for “0 TFA” claims in an actual e-commerce market, complementing survey-based literature [25–27]. Second, we offer a structural measurement of the fat health halo. While emerging literature suggests digital alerts shift WTP [27], few studies have isolated the specific role of hero-image claims in driving the indirect distortion of nutrient valuation. By modeling the interaction between the claim and total fat utility, we distinguish the direct premium for being “0 TFA” from the induced distortion in fat valuation—a direct test of the halo hypothesis. Third, we derive policy implications, suggesting that if hero-image claims command high premiums while increasing the willingness to pay for high total fat, regulatory regimes may need to address how voluntary claims are displayed digitally to prevent unintended health halos.

2. Relevant Studies

Price premia and willingness to pay (WTP) for credence attributes—such as "healthy," "organic" , or "sustainable"—have been widely analysed in food economics [28–30]. To situate the specific role of "0 TFA" claims within online bakery markets, we organize the relevant literature into two closely related strands: (i) consumer demand and WTP for voluntary food attributes across both offline and digital retail environments; and (ii) behavioural mechanisms, particularly health halos, whereby nutrition and health claims reshape perceived product quality and nutrient valuation. Accordingly, Section 2.1 synthesizes the evidence on WTP for voluntary attributes, with a specific focus on "0 TFA" claims, while Section 2.2 reviews the health halo literature to clarify how our setting extends these insights and to identify the critical gaps motivating our empirical analysis.

2.1. Consumer Demand and Willingness to Pay for Voluntary Food Attributes

General evidence shows that providing nutrition information—whether through back-of-pack panels or front-of-pack logos—can shift market shares toward healthier options and incentivize firms to improve product quality [31].

A growing body of research uses e-commerce platforms as natural laboratories to study these valuations, showing that digital product pages can amplify the value of voluntary labels. For instance, significant implicit prices for organic and origin certifications were identified in extra-virgin olive oils sold in the Italian online market [32], while substantial premiums for organic and origin attributes were reported for dried pasta sold via e-commerce [33]. In the context of China, rice bearing sustainability labels—such as "Green Food" or "Organic Food"—was shown to command price premiums of approximately 40–50% [25]. Furthermore, "Green Food" labels were found to earn higher premiums online compared to offline channels, underscoring the amplifying role of the digital shelf [26]. Beyond food safety labeling, empirical studies indicate that greater transparency regarding environmental or origin attributes increases purchase intentions [34,35]. Regarding logistics and carbon disclosure, a willingness to pay premiums for green express packaging has been documented among Chinese consumers [36,37]. Similarly, in the European market, CO_2 emission labels in German online shops have been observed to increase stated willingness to pay [38].

Regarding Trans-Fatty Acids (TFA) specifically, existing research focuses primarily on regulatory effects and offline valuation. Evidence from high-income countries indicates that mandatory declaration and subsequent bans were highly effective: Substantial reductions in TFA content following regulatory interventions have been documented [1,4]. Using scanner data, it was shown that mandatory TFA labelling shifted consumption shares toward reformulated products [39], while manufacturers were frequently observed using "no trans fats" claims to signal compliance and quality [40]. In terms of explicit monetary valuation in brick-and-mortar settings, a hedonic model applied to the U.S. biscuit market revealed that a voluntary "0 TFA" label commands an implicit price premium of approximately 17.6% [41]. Similarly, positive implicit prices for trans fat claims (meaning low or no trans fat) have been identified in soup products [42]. Complementing these price analyses, a supermarket shelf-label experiment demonstrated that adding "0 TFA" labels significantly increases sales volume, indicating a strong revealed preference for this attribute in offline retail [43].

However, a critical gap remains. The regulatory literature typically examines aggregate shifts in product mix or nutrient intake and rarely captures demand-side incentives at the individual product level or considers institutional settings like China, where consumers rely heavily on voluntary claims. At the same time, e-commerce studies confirm high WTP for attributes related to sustainability, traceability, or packaging, but current estimates of WTP for "0 TFA" claims are confined to traditional offline retail environments.

Research on the valuation of TFA-related claims in online markets therefore remains absent. Our study fills this gap by using real market data from China's digital marketplace to provide the first revealed-preference estimates of the "trans fat-free" premium in an e-commerce setting.

2.2. Health Halos and Behavioural Mechanisms

The second strand documents the behavioural mechanisms that might drive such valuations. Extensive experimental evidence shows that favourable messages about a single attribute can generate a "health halo", leading consumers to judge the overall product more positively. For instance, "low-fat" or "high-protein" labels have been shown to increase perceived healthfulness and intended portion sizes, even when the product remains high in total energy or sugar [12,13].

Regarding TFA specifically, it has been experimentally demonstrated that front-of-pack claims—including "0 TFA" (e.g., on chicken nuggets)—significantly increase the probability of pre-adolescent children selecting energy-dense, nutrient-poor foods over healthier alternatives [44]. Complementing this observation, it was found that even with mandatory back-of-pack disclosure, the presence of a "low trans-fat" claim significantly enhances perceived nutrition and reduces disease risk assessments [45]. These results align with the theory of self-licensing, where a "virtuous" cue grants individuals a "license to sin," thereby facilitating indulgent choices [14].

However, existing research provides only a partial view of this halo effect, limited primarily to offline or hypothetical experimental settings. These studies typically examine how claims change choice probabilities or stated risk perceptions for processed foods (e.g., snacks, nuggets, or crackers), rather than quantifying how they reshape the monetary value consumers place on nutrients themselves. Critically, empirical evidence regarding this 'health halo' in the actual online marketplace—where visual cues like hero images are dominant—remains absent. Our study fills this gap by quantifying how a "0 TFA" claim changes the marginal willingness to pay (WTP) for total fat in a staple food category—bread—thereby turning the health halo into an explicit trade-off between fat content and price. Specifically, we examine whether the presence of a "0 TFA" claim positively shifts the monetary valuation consumers assign to total fat content. By using actual transaction data from a major online platform, we provide revealed-preference evidence of this "fat health halo" in a real-world digital market."

3. Method

3.1 Rationale: Revealed Preference Approach

While discrete choice experiments (DCEs) are widely used in food research to elicit consumer preferences, they often rely on stated preference data from surveys, which can be subject to hypothetical bias [28,29,46–49]. In this study, we adopt a revealed-preference approach, estimating a discrete choice demand model using aggregate market shares from a live-streaming e-commerce platform [22–24,50,51]. Unlike survey-based methods, this approach infers Willingness-to-Pay (WTP) from actual purchasing decisions made under real budget constraints and market prices [30,49]. By relying on realized transactions, we avoid the 'attitude-behavior gap' often observed in ethical or healthy food consumption, providing a strictly behavioral measure of the claim's value [52,53]. Following the framework of discrete choice models using market-level data[23], we utilize data aggregated to market shares—structured as a brand-market panel—to estimate discrete choice parameters without relying on individual-level purchase records, a strategy widely applied in food economics WTP research [50,54]. By observing daily variations in sales and prices, we can recover consumer valuation for nutritional attributes while explicitly modeling the "outside option" (consumers who browse but do not buy), thereby capturing the

substitution between purchasing a specific bread product and exiting the category entirely [22,51].

3.2 Utility Specification (Baseline Model)

We model consumer choice using a random utility framework [51]. We begin with a baseline fixed-coefficient specification, which assumes that consumer preferences are homogeneous. The indirect utility of consumer i purchasing product j in market (day) t is explicitly specified as Equation (1):

$$u_{ijt} = \alpha p_{jt} + \beta_1 \text{ZeroTFA}_j + \beta_2 \text{Fat}_j + \beta_3 (\text{ZeroTFA}_j \times \text{Fat}_j) + Control'_{jt}\gamma + \xi_{jt} + \varepsilon_{ijt} \quad (1)$$

In Equation (1), the key variables of interest are explicitly defined. p_{jt} denotes the retail price per 100g, while ZeroTFA_j serves as a binary indicator equal to one if the product displays a voluntary "0 TFA" hero-image claim. We also include Fat_j, representing the demeaned total fat content per 100g, along with its interaction with the claim, $\text{ZeroTFA}_j \times \text{Fat}_j$. This interaction term allows us to test for a "health halo" effect; specifically, a positive coefficient β_3 would imply that the presence of the "0 TFA" claim increases the consumer's marginal utility associated with higher fat content. [31,49,55–58].

To prevent omitted variable bias, the vector $Control_{jt}$ incorporates a comprehensive set of product and market attributes. First, we control for other physical attributes, including the "Whole Wheat" (Quanmai) claim and key nutrient contents (Energy, Carbohydrates, Sodium, and Protein). Including these ensures that the estimated effect of the "0 TFA" claim reflects the value of the label itself rather than correlated nutritional qualities [46,49,56]. Second, we account for marketing exposures by including dummy variables for daily Short Video (Video) and Live-stream (Live) promotions. Given that sales in live-streaming e-commerce are heavily driven by algorithmic traffic, controlling for these exposures is crucial to separate demand driven by intrinsic product preference from demand driven by platform visibility [59]. Finally, we include fixed effects for Manufacturer, Season, Origin, and Year to absorb unobserved heterogeneity arising from brand loyalty, seasonal demand fluctuations, and region-of-origin preferences. The term ξ_{jt} represents unobserved product quality, and ε_{ijt} is the idiosyncratic error term.

We assume that the idiosyncratic error term ε_{ijt} follows an independent and identically distributed (i.i.d.) Type I extreme value distribution. Let δ_{jt} denote the mean utility of product j in market t, which captures the deterministic component of consumer utility (i.e., $\delta_{jt} = u_{ijt} - \varepsilon_{ijt}$). The individual probability of choosing product j is given by P_{ijt} [51]:

$$P_{ijt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}$$

3.3 From Individual Choice to Aggregate Market Shares

Following the extensive literature on estimating the WTP for food attributes using discrete choice models with aggregate data [24,50,54], we link individual utility to our aggregate sales data.

In practice, we construct a consistent predictor of actual market shares by aggregating these individual probabilities over the population: $\widehat{s_{jt}} = \frac{1}{M_t}\sum_{i=1}^{M_t} P_{ijt}$. In large samples, the model requires that the predicted shares equal the actual observed shares:

$$s_{jt}(\delta_{jt}) = S_{jt} \equiv \frac{q_{jt}}{M_t}$$

where S_{jt} represents the observed market share (calculated as daily sales volume q_{jt} divided by potential market size M_t, proxied by category-level page views). In the

baseline fixed-coefficient model, the predicted share s_{jt} is analytically equivalent to the closed-form Logit expression:

$$s_{jt}(\delta_{jt}) = P_{ijt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}$$

Similarly, the outside option share s_{0t} corresponds to the residual probability of consumers choosing none of the available inside products [22,51].

According to the literature on estimating discrete choice models using aggregate market shares, markets are typically defined either by a combination of geography and time (e.g., city-quarters) or by a single temporal dimension when product prices do not vary across regions [22–24,60]. In our context, the e-commerce platform operates as a unified national marketplace, allowing consumers throughout China to purchase the exact same products at identical prices. Consequently, we define the market purely along the temporal dimension, treating each day as a distinct market. Furthermore, following standard conventions for defining the potential market size (M_t) in digital retail, we utilize the platform's daily total page views for the bread category as the proxy for the potential market, thereby treating browsing as purchase consideration where visits without transactions constitute the outside option [61–63].

3.4 Linear Estimation Strategy

Based on the framework established for estimating discrete choice models using aggregate product–market data, the market share system is inverted to obtain a linear regression equation [22,24,50]. By normalizing the mean utility of the outside option to zero, Equation (2) allows us to derive the following linear estimating equation:

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt} = \alpha p_{jt} + \beta_1 \text{ZeroTFA}_j + \beta_2 \text{Fat}_j + \beta_3(\text{ZeroTFA}_j \times \text{Fat}_j) + Control'_{jt}\gamma + \xi_{jt} \quad (2)$$

This linear equation represents the closed-form transformation for the standard logit model with fixed coefficients. It allows us to estimate parameters using linear instrumental variable techniques [22,24,51]. The dependent variable becomes the log-odds of purchase relative to the outside option[1].

A key challenge is price endogeneity: retail prices (p_{jt}) are likely correlated with unobserved demand shocks ξ_{jt}. To address this, we estimate the equation using Two-Stage Least Squares (2SLS), instrumenting the retail price with the weekly wholesale price of wheat and sugar. As a cost-shifter, wheat price variations drive retail prices but are exogenous to daily, platform-specific demand shocks [24,54].

In the actual estimation, we mean-centered the total fat variable by subtracting the sample mean. This ensures that the coefficient for the "0 TFA" claim captures consumer response at the average fat level rather than at a hypothetical zero-fat baseline, enhancing practical interpretability without affecting other parameter estimates in line with standard practice in food demand models with interactions [64,65].

Finally, the Willingness-to-Pay (WTP) for the "0 TFA" claim at the average fat level is calculated based on the baseline estimates as:

$$\text{WTP}_{\text{ZeroTFA}} = -\frac{\beta_1}{\alpha} \quad (3)$$

This provides a monetary valuation of the claim for the average consumer [29,49,51].

Furthermore, to quantify the "Health Halo" effect—defined as the distortion in consumer valuation of total fat induced by the claim—we derive the shift in the Marginal

1. This method is referred to as the Berry inversion[22].

Willingness-to-Pay (MWTP) for fat content. We do not impose *a priori* assumptions on the sign of the baseline preference for fat (β_2), acknowledging that consumers may value fat positively for taste or negatively for health concerns.

The marginal utility of total fat is given by $\partial u / \partial Fat = \beta_2 + \beta_3 \cdot ZeroTFA$. Accordingly, the monetary valuation of fat shifts based on the presence of the claim:

$$MWTP_{Fat} = -\frac{\beta_2 + \beta_3 \cdot ZeroTFA}{\alpha} \quad (4)$$

The Halo Valuation (WTP_{Halo}) is isolated as the difference between these two states, representing the specific change in the marginal value of fat attributed solely to the "0 TFA" claim:

$$WTP_{Halo} = MWTP_{Fat|Claim} - MWTP_{Fat|NoClaim} = -\frac{\beta_3}{\alpha}$$

This metric captures the monetary magnitude of the halo effect. A positive interaction coefficient ($\beta_3 > 0$)—given a negative price coefficient ($\alpha < 0$)—implies a positive WTP_{Halo}. This indicates that the claim systematically increases the perceived value of fat (or reduces the penalty associated with it), thereby quantifying how the visual cue distorts nutrient valuation in the digital marketplace.

3.5 Extended Specification: Modeling Preference Heterogeneity

The baseline specification in Equation (1) assumes that consumers' valuation of the "0 TFA" claim is homogeneous. However, health consciousness often varies across the population. To address this limitation and test the robustness of our results, we introduce an alternative Random Coefficients Logit (RC-Logit) model that explicitly accounts for preference heterogeneity [23,24,51].

The utility function for this comparative random coefficients model is specified as:

$$u_{ijt} = \alpha p_{jt} + \beta_{i,\text{ZeroTFA}}\text{ZeroTFA}_j + \beta_2\text{Fat}_j + \beta_3(\text{ZeroTFA}_j \times \text{Fat}_j) + Control'_{jt}\gamma + \xi_{jt} + \varepsilon_{ijt} \quad (5)$$

In Equation (5), the coefficient for the "0 TFA" claim varies across consumers: $\beta_{i,\text{ZeroTFA}} = \overline{\beta_{\text{ZeroTFA}}} + \sigma_{\text{ZeroTFA}}\nu_i$, where σ_{ZeroTFA} captures the standard deviation of tastes. In this random coefficient model, we estimate the average WTP using Equation (3) and (4), simply substituting the mean coefficient $\beta_{ZeroTFA}$ for β_1. A significant σ implies that consumers are heterogeneous: some strongly value the claim, while others do not. Estimating random coefficient models using aggregate brand–market panel data typically employs Generalized Method of Moments (GMM) and simulation methods, as this specification lacks a closed-form linear solution. [23,24,51].

We compare the results from these models across three specifications to ensure robustness[2]:

Logit: Baseline fixed-coefficient model [Eq. (1)] (ignoring endogeneity).

IV-Logit: Baseline fixed-coefficient model [Eq. (1)] (with IV).

RC IV-Logit[3]: Full structural model [Eq. (5)] (heterogeneity + IV).

4. Data and Descriptive Statistics

4.1. Data Source and Sample Selection

2. Consistent with previous studies (e.g., Berry et al., 1995; Nevo, 2001), we omit the random coefficients model with exogenous prices, given that identification with aggregate data relies on instrumental variables.

3. This modeling framework is frequently termed the BLP approach and is extensively used in the estimation of food demand [24,50,54].

We constructed a comprehensive dataset from a major live-streaming e-commerce platform in China, covering the period from September 1, 2024, to August 14, 2025. Ranking among the top three e-commerce marketplaces in China in 2024, this platform serves as a critical barometer for the digital food retail market. To ensure the representativeness and data quality of our analysis, our sample selection followed a strict, systematic process rather than random sampling. We focused on the broader bread category and applied two inclusion criteria: (1) Market Significance: We selected only top-selling SKUs with cumulative sales exceeding 100,000 units from the top-80 sellers, ensuring the sample captures mainstream market preferences; (2) Data Quality: We restricted the sample to products that provided standardized, publicly accessible nutritional information in their online details. Based on these criteria, we tracked 59 distinct high-selling bread products, which together accounted for more than half of all bread-category sales on the platform in the first half of 2025. This coverage makes our sample at least as representative as many comparable single-category studies that focus on leading or top-selling products within a category [66–69].

Structured as a product-date panel, our dataset tracks these specific items on a daily basis. However, due to real-world inventory fluctuations where products might be unavailable on certain days, the dataset constitutes an unbalanced panel with a total of 15,499 daily observations.

Crucially, the disclosure of nutritional content in our sample exhibits a distinct asymmetry. While all products publicly display the mandatory Nutrition Information Panel (NIP)—covering energy, total fat, protein, carbohydrate, and sodium—in the product details section, the disclosure of Trans-Fatty Acids (TFA) is strictly voluntary. Only products explicitly carrying the "0 TFA" claim in their hero image disclose their TFA status (indicating zero content). Figure 1 illustrates a typical example of this "0 TFA" claim embedded within the product's hero image. For all other products, TFA information is completely absent from both the hero image and the detailed NIP.

Figure 1. Example of a “0 TFA” claim embedded in a product’s hero image. To avoid potential copyright infringement and brand identification, the original product image has been blurred, highlighting only the position and content of the claim.

Finally, to address price endogeneity, we incorporated the corresponding weekly wholesale prices of wheat and granulated sugar as instrumental variables. Using raw material prices as instrumental variables is a standard practice to address price endogeneity

in food demand studies[24,54]. These data were sourced from the Qianzhan Database (Qianzhan Industrial Research Institute, Shenzhen, China), based on official statistics from the National Bureau of Statistics and Chinese customs.

4.2. Descriptive Statistics and Nutritional Uniformity

Table 1 presents the descriptive statistics for the full panel dataset used in our empirical estimation ($N = 15{,}499$). The average price of the sampled bread products is 3.623 CNY per 100g ($SD = 1.937$). Marketing intensity on the platform is substantial, with an average of 23.373 short videos and 19.276 live-streaming sessions linked to product listings daily. Regarding the visual attributes, approximately 63% of the daily observations feature the "0 TFA" claim in their hero image ($Mean = 0.633$). The nutritional profile of the full sample shows an average energy content of 1,354.135 kJ/100g and a total fat content of 9.737g/100g.

Table 1. Descriptive statistics

Variable	Unit	Mean	SD	Min	Max
Price	CNY/100g	3.623	1.937	0.796	8.069
Short Video		23.373	36.808	0.000	388.000
Live Streaming		19.276	31.866	0.000	352.000
0 TFA Claim	Dummy (0/1)	0.633	0.482	0.000	1.000
Energy	kJ	1354.135	194.811	807.353	1802.000
Total Fat	g/100g	9.737	5.057	0.000	19.600
Protein	g/100g	9.184	1.908	5.200	16.200
Carbohydrate	g/100g	49.271	7.181	32.900	78.100
Sodium	mg/100g	256.410	119.332	122.000	781.000
Observations	15,499				

Note: Data represents original (non-demeaned) values. Nutritional attributes are time-invariant for each product; statistics reflect the distribution across daily observations.

A critical identification challenge arises from the invariant nature of the claims within our sample, which consists of 36 products that consistently displayed the "0 TFA" claim and 23 that never did throughout the observation period. Given this lack of within-product variation, we face the concern that the "0 TFA" claim may merely act as a proxy for unobserved product heterogeneity. Specifically, if the labelled group inherently possesses superior overall nutritional quality (e.g., lower total fat or calories) compared to the non-labelled group, it becomes difficult to disentangle the marginal effect of the claim itself from the fundamental nutritional attributes of the products. To address this, we compared the macronutrient profiles of products with and without the claim at the SKU level ($N = 59$). As shown in Table 2, the nutritional composition is highly consistent between the two groups. Products with the "0 TFA" claim have an average energy content of 1,368.522 kJ/100g, while those without the claim average 1,320.462 kJ/100g. Similarly, the total fat content shows no substantial difference (9.399 g vs. 10.017 g per 100g). Independent t-tests confirm that there are no statistically significant differences in energy, total fat, protein, carbohydrate, or sodium content between the two groups ($p > 0.1$).

Table 2. Comparison of nutritional composition between bread products with and without voluntary "0 TFA" claims

Nutritional Attribute (per 100g)	With "0 TFA" Claim (n = 36)	Without Claim (n = 23)	Mean Difference	T-Value	P-Value
Energy (kJ)	1368.522 (210.563)	1320.462 (184.097)	48.060	0.897	0.374
Total Fat (g)	9.399	10.017	-0.618	-0.450	0.655

	(5.105)	(5.212)			
Protein (g)	9.157	9.093	0.064	0.126	0.900
	(1.913)	(1.916)			
Carbohydrate	50.506	47.633	2.873	1.429	0.159
(g)	(8.597)	(5.431)			
Sodium (mg)	261.362	234.917	26.445	0.863	0.392
	(122.154)	(102.060)			
Observations	59				

This lack of observable nutritional difference implies that, on average, consumer Willingness to Pay (WTP) for the labeled products is derived principally from the "trans fat-free" attribute itself (and its associated safety signaling), rather than from correlated nutritional benefits such as lower calorie or fat content. Nevertheless, to ensure robust estimation, we will strictly control for all these nutritional variables, along with prices and marketing intensity, in our subsequent regression analysis.

5. Results

The estimation results for the structural parameters of consumer utility are presented in Table 3, while the derived monetary valuations (WTP) are summarized in Table 4. Across all analyses, we compare the Logit, IV-Logit, and Random Coefficients (RC) IV-Logit specifications. Diagnostic tests further validate our identification strategy, yielding a first-stage F-statistic of 19.21 that rules out weak identification and a non-significant Hansen J statistic ($p = 0.41$) supporting the exogeneity of the instruments. A defining feature of our results is the high degree of consistency in the direction and statistical significance of key parameters across all three models, providing strong evidence for the robustness of our conclusions.

5.1. Preference Estimates and Model Robustness

Table 3 reports the utility coefficients. First, regarding price sensitivity, the coefficient for price (p_{jt}) is consistently negative and statistically significant at the 1% level across all three specifications ($p < 0.01$). The magnitude of the price coefficient increases substantially from -0.110 in the Logit model to -0.346 in the IV-Logit model, and to -0.354 in the RC IV-Logit model. This shift confirms the successful correction of endogeneity bias using the wholesale price instruments.

Second, the structural preference for the "0 TFA" claim is robust. The coefficient for the "0 TFA" dummy (β_1 in Logit or IV-Logit, $\beta_{ZeroTFA}$ in RC IV-Logit) is positive and highly significant across all models ($p < 0.01$ or $p < 0.05$). Notably, the use of instrumental variables substantially alters the magnitude of the estimate: the coefficient rises from 0.104 in the standard Logit model to approximately 0.3 in both IV-based specifications, indicating that price endogeneity biases the estimation of other preference parameters. Nevertheless, the overall results remain robust. This consistency demonstrates that consumers derive significantly higher utility from products displaying the hero-image claim.

Third, the coefficient for total fat content (β_2) is consistently positive and statistically significant across all specifications ($\beta_2 \approx 0.036,\ p < 0.01$). This indicates that consumers generally exhibit a positive baseline preference for the taste and texture associated with fat, a finding that aligns with general sensory expectations and remains robust regardless of the model used. This aligns with previous literature suggesting that since low-fat content can compromise hedonic enjoyment, consumers are sometimes willing to pay a premium for higher fat to preserve palatability, despite the associated health risks[70–74].

Furthermore, regarding heterogeneity, the RC IV-Logit model (Model 3) identifies a statistically significant standard deviation for the "0 TFA" coefficient ($\sigma = 0.082,\ p <$

0.1). While this indicates that consumer sensitivity to the claim varies across the population, the mean coefficient in the RC IV-Logit model remains nearly identical to the fixed-coefficient IV-Logit estimate. This stability confirms that our baseline average preference estimates are not driven by model specification artifacts.

Table 3. Demand for sliced bread: price sensitivity and "0 TFA" fat halo across specifications

Variable	Logit	IV-Logit	RC IV-Logit
Price (per 100 g)	-0.110*** (0.009)	-0.346** (0.166)	-0.354** (0.166)
"0 TFA"	0.104*** (0.029)	0.317** (0.154)	0.321** (0.157)
Fat	0.034*** (0.008)	0.036*** (0.008)	0.037*** (0.008)
"0 TFA" × Fat	0.029*** (0.006)	0.045*** (0.013)	0.045*** (0.013)
σ("0 TFA")	/	/	0.082* (0.046)
Controls[1]	Yes	Yes	Yes
Price IV		Yes	Yes
Observations	15,499	15,499	15,499

5.2. Willingness to Pay for the "Trans Fat-Free" Claim

We derived the monetary valuation of the "0 TFA" claim by normalizing the utility parameters by the marginal utility of income. As shown in Table 4, the estimated WTP exhibits remarkable robustness.

Despite the variation in raw coefficients between different models, the ratio of the "0 TFA" preference to the price coefficient remains stable. The WTP for the claim is estimated at 0.945 CNY in the Logit model, 0.916 CNY in the IV-Logit model, and 0.907 CNY in the RC IV-Logit model. The tight range of these estimates confirms that the valuation is structurally sound.

Taking the RC IV-Logit model as an example, the WTP of 0.907 CNY represents approximately 25% of the average product price (3.623 CNY/100g). This substantial premium indicates that the "0 TFA" claim acts as a high-value signal in the digital marketplace, commanding a price markup that is consistently detected across all modeling approaches.

Table 4. Willingness to pay (WTP) for "0 TFA" claim and total fat (per 100 g)

WTP	Logit	IV Logit	RC IV-Logit
"0 TFA"	0.945	0.916	0.907
Fat (No Claim)	0.309	0.104	0.105
Fat (With Claim)	0.573	0.234	0.232
Halo Valuation (ΔWTP for fat)	0.264	0.130	0.127

5.3. The "Health Halo" Effect

The central finding of this study is the identification of a significant "Health Halo," captured by the interaction term between the voluntary claim and total fat content (ZeroTFA × Fat). As reported in Table 3, the coefficient for this interaction (β_3) is consistently positive and statistically significant at the 1% level across all three specifications. Specifically, in the IV-Logit and RC IV-Logit models, the interaction coefficient stabilizes at approximately 0.045, confirming that the presence of the "0 TFA" claim systematically increases the marginal utility consumers derive from total fat, independent of preference heterogeneity. Although the Logit model yields a lower estimate of 0.029—reflecting the

bias from price endogeneity—it remains significantly positive, further confirming the robustness of the result.

Table 4 translates these utility parameters into monetary terms, revealing the magnitude of this distortion. In the baseline state (without the claim), consumers exhibit a modest positive valuation for fat, with a Marginal Willingness to Pay (MWTP) of 0.105 CNY per 100g in the RC IV-Logit model. However, when the "0 TFA" hero-image claim is present, this valuation more than doubles, rising to 0.232 CNY per 100g. This pattern of increased valuation is consistently observed across the alternative Logit and IV-Logit specifications, confirming the robustness of the result.

The difference between these two states represents the "Halo Valuation," isolated at 0.130 CNY in the IV-Logit model and 0.127 CNY in the RC IV-Logit model. This corresponds to an approximate 121% increase in the valuation of fat relative to the no-claim baseline. In the Logit model, this difference reached 0.264 CNY. The divergence of this estimate from the IV-based specifications is primarily attributable to uncorrected price endogeneity, yet this difference does not alter the main conclusion. These results provide robust revealed-preference evidence that the "0 TFA" label functions as more than just a safety signal; it acts as a value multiplier for total fat, effectively reducing the perceived health cost or increasing the sensory value of fat in the eyes of the consumer.

6. Discussion

6.1. Discussion of Main Results

This study used real brand-market share data from a major live-streaming e-commerce platform to quantify how voluntary “0 TFA” hero-image claims affect consumer valuation of bread and its fat content. By estimating a discrete choice model with price endogeneity corrected through cost-based instruments, we provide two main conclusions.

First, consumers place a sizeable and robust monetary value on the “0 TFA” claim itself. Across all specifications, the estimated WTP for the claim is close to 0.9 CNY per 100 g, with the RC IV-Logit model indicating a premium of 0.907 CNY. This corresponds to roughly one quarter of the average unit price in our sample, suggesting that "0 TFA" has become a salient and highly valued credence attribute in the digital bakery market, even in the absence of mandatory TFA disclosure on the package.

Second, and most importantly, we document a pronounced “fat health halo” effect in a real-world digital marketplace. The presence of the “0 TFA” hero-image claim markedly increases the marginal WTP for total fat. In the RC IV-Logit model, the WTP for fat rises from 0.105 CNY per 100 g in the absence of the claim to 0.232 CNY when the claim is present, implying a halo valuation of 0.127 CNY. This represents an increase of about 121% relative to the baseline valuation. Thus, the “0 TFA” label does not simply add an independent premium; it further amplifies the existing positive preference consumers hold for fat content. The random-coefficients specification confirms that these patterns are not driven by model artifacts and that there is meaningful heterogeneity in how different consumers respond to the claim.

Taken together, these findings show that voluntary “0 TFA” hero-image claims on digital shelves generate a substantial price premium and, at the same time, distort the valuation of fat in a direction that is potentially misaligned with public health goals. The results demonstrate that the push to eliminate industrial TFA, if implemented through voluntary digital claims, may unintentionally reinforce consumer incentives to purchase high-fat products within the same category.

6.2. Policy Implications

Our findings carry significant implications for nutrition policy and the governance of digital food environments.

First, the results suggest that TFA-specific claims must be presented within a more balanced nutritional context [75–77]. Our data reveals that "0 TFA" claims command a substantial price premium and induce a positive halo effect regarding fat valuation. Regulatory frameworks that focus exclusively on industrial TFA, without ensuring transparency regarding other nutrients, may inadvertently incentivize marketing strategies that highlight the absence of TFA while obscuring high total fat content [75,76]. Consequently, regulators should consider mandating that whenever a "0 TFA" claim is utilized, the total fat content must be prominently highlighted or displayed in close proximity [77]. This would prevent consumers from being misled by a single safety attribute and ensure they are presented with a holistic view of the product's nutritional profile [76]. Second, the digital food environment demands specific guidelines regarding the display of voluntary claims [78,79]. On live-streaming and e-commerce platforms, the "0 TFA" statement is primarily communicated through the hero image—the first visual cue consumers encounter [78]. Our findings indicate that this visually prominent cue can significantly reshape how consumers value product attributes. Therefore, regulatory authorities could extend front-of-pack (FOP) labelling principles to "digital FOP" formats by requiring that nutrient-related claims in hero images be accompanied by concise, standardized nutrient information (e.g., a summary indicator) to prevent the selective emphasis of positive attributes [78]. Third, platforms and retailers themselves possess the mechanisms to mitigate misleading halo effects [78,79]. Given that search ranking algorithms and recommendation systems determine which hero images are most visible, platforms could prioritize products that combine "0 TFA" claims with healthier overall nutritional profiles when users search for health-related keywords. Alternatively, they could implement verification procedures to reduce the risk of "0 TFA" being used strategically to confuse consumers [78]. Finally, consumer education remains indispensable [76,77]. Our estimates reveal that many shoppers are willing to pay a sizeable premium for "0 TFA" while paradoxically valuing higher fat content when the claim is present. Public health campaigns should emphasize that "0 TFA" is not synonymous with "low fat" or "overall healthy," thereby helping consumers interpret claims within the context of the full nutrient profile [75–77].

6.3 Limitations and Future Work

This study has several limitations that suggest avenues for further work.

First, the analysis focuses on a single product category—bread—in one large Chinese live-streaming e-commerce platform. While bread is a relevant staple with substantial sales volume, the magnitude and direction of health halos may differ in snack foods, confectionery, or ready-to-eat meals, as well as across platforms and countries. Future research could replicate the approach in other categories and jurisdictions.

Second, our data are aggregated at the brand-market level and do not include individual-level socio-demographic or health information. As a result, we can document average and model-based heterogeneous responses, but we cannot directly relate WTP or halo effects to consumer characteristics such as age, income, or nutrition knowledge. Combining demand models with user-level clickstream data would allow a more fine-grained analysis.

Third, although the use of cost-based instruments reduces concerns about price endogeneity, our estimates are still based on observational data. Future studies could integrate experimental designs—such as A/B tests that manipulate hero-image claims—to provide even stronger causal evidence.

Finally, our analysis focuses specifically on the interaction between "0 TFA" and total fat. Extending the model to multiple simultaneous claims (e.g., "low sugar," "high fiber") would help unpack how complex digital labels shape nutrient valuation in an evolving regulatory context.

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Abbreviations

The following abbreviations are used in this manuscript:

TFA	Trans-Fatty Acids
IV	Instrumental Variable
RC	Random Coefficient
SKU	Stock Keeping Unit

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